

Research Article



Retrospective study of fracture survival in endodontically treated molars: the effect of single-unit crowns versus direct-resin composite restorations

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Conflict of Interest

No potential conflict of interest relevant to this article was reported.

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INTRODUCTION

The success of endodontic treatment is strongly affected by the quality of root canal treatment and coronal restoration [1-3]. In endodontically treated teeth (ETT), coronal restorations provide a coronal seal and protect the remaining tooth structure from fracture

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[4,5]. Fracture of ETT usually occurs when ETT are not immediately restored, resulting in coronal microbial leakage or an unrestorable fracture [6-8].

The amount of remaining tooth structure affects the fracture resistance of ETT [4,9-12]. Tooth structure loss, quantified in terms of the number of sides, has been reported to be a significant factor associated with reduced tooth stiffness and an increased risk of fracture in posterior ETT [13-15]. Loss of marginal ridge(s) decreases tooth stiffness, increases cuspal flexure and, consequently, leads to a risk of tooth fracture [11,12,16].

Previous retrospective studies have indicated that placement of a full-coverage restoration significantly improved the overall survival rate of posterior ETT after fractures [8,13,14]. Nevertheless, a recent systematic review reported that the long-term success of resin composite restorations in posterior ETT depended on the amount of tooth structure loss [5]. A recent clinical study demonstrated a high survival rate of posterior ETT with composite fillings that had different amounts of tooth surface loss, but the results did not report the survival separately by tooth type [14]. Retrospective studies have presented higher survival rates from fracture for premolar ETT with mild to moderate tooth structure loss that were restored with resin composite restorations than for those restored with full-coverage crowns [13,17,18]. However, the results of these studies are limited to only premolar ETT.

A previous study showed an acceptable survival rate of molar ETT with an adequate amount of remaining tooth structure that were restored with resin composite in a short-term period. However, that study did not compare the survival with ETT restored using crown restorations [19]. To date, limited research has compared the fracture survival rate between direct resin composite and crown restorations in molar ETT. Therefore, the purpose of this study was to compare the fracture survival rates of molar ETT restored with direct resin composites or crowns, using a retrospective cohort design. In addition, the potential risk factors associated with fracture survival were identified.

MATERIALS AND METHODS

The protocol of this retrospective study was approved by the Institutional Ethics Committee of Mahidol University (MU-DT/PY-IRB 2020/019.3003) and Rangsit University (DPE. No. RSUERB2020-020). The dental records of patients who received nonsurgical endodontic and restorative treatments and presented at a recall visit from 2015 to 2019 were collected. Some of the ETT molars were selected from our previous study, with a longer follow-up period, and some new ETT were included [13]. The data of the ETT were reviewed and included in the study if the teeth satisfied the following criteria.

The inclusion criteria for ETT were 1) first and second molars with mature root formation; 2) teeth restored with single-unit full-coverage crowns or direct resin composite restorations; 3) a recall examination period of at least 12 months; and 4) adequate data regarding treatment procedures, clinical records, and radiographic records. The exclusion criteria were 1) teeth extracted for other reasons, not due to unrestorable fracture; 2) chronic periodontitis with bone loss of more than half of the root length; 3) preoperative cracks or radicular fracture; 4) history of surgical root canal treatment, hemisection, or root amputation; and 5) fixed orthodontic appliances.

Endodontic and restorative procedures

Root canal treatments were performed by undergraduates, postgraduates, or endodontists under rubber dam isolation. Root canal cleaning and shaping were performed with the crown-down technique using 0.02 taper hand stainless steel and/or 0.04/0.06 taper rotary Ni-Ti files with 2.5% sodium hypochlorite and 17% EDTA irrigants. After calcium hydroxide intracanal medication, the prepared root canals were then obturated with gutta-percha cones and root canal sealer (zinc oxide, epoxy resin, or calcium silicate-based sealer), using one of the following root canal obturation methods: lateral compaction, vertical compaction, or the cold hydraulic (sealer-based) technique.

Coronal restorations were performed by undergraduates, postgraduates, or prosthodontists. Full-coverage crown restorations (with core build-up or post placement) were commonly planned as a final restoration. Direct resin composite restorations were placed as a long-term, interim restoration in cases with a questionable prognosis in terms of endodontics or periodontics, or those on a waiting list for crown restorations. Resin composite was occasionally used as a permanent restoration for patients who could not afford the cost of a crown, or for ETT with tooth structure loss limited to the occlusal surface only.

In cases that were restored with crown restorations, a post was only indicated when the retention from the remaining tooth structure for core build-up was not adequate. Cast metal posts or prefabricated fiber posts (D.T. LIGHT-POSTS (BISCO Inc., Schaumburg, IL, USA) or FRC Postec Plus (Ivoclar Vivadent AG, Schaan, Liechtenstein) were cemented into the prepared root canals with adhesive resin-based cement (Panavia F 2.0, Kuraray Noritake Dental Inc., Tokyo, Japan) or core build-up material (MultiCore Flow, Ivoclar Vivadent AG).

In ETT restored with resin composite restorations, resin composite was placed without the use of a post. The access cavity was prepared using Cavit (CAVITON, GC Corp., Tokyo, Japan), or glass-ionomer cement liner (Vitrebond, 3M ESPE, St. Paul, MN, USA; or GC Fuji VII, GC Corp.) applied at a thickness of 1-2 mm. Resin composite (Z250 or Z350, 3M ESPE) was placed and bonded with etch-and-rinse adhesive (Adper Single Bond 2, 3M ESPE), or self-etch adhesive (Clearfil SE Bond, Kuraray Noritake Dental Inc.).

Data collection and assessment criteria

An overview of the methodology is presented in **Figure 1**. From clinical and radiographic records, the following information was collected: age, sex, recall period (months), tooth type, tooth location (upper or lower), coronal restoration (crown or resin composite), opposing dentition (natural tooth, fixed prosthesis, removable prosthesis, or none), and number of adjacent teeth. For ETT restored with resin composite restorations, the amount of tooth surface loss was identified from dental records. For ETT restored with full-coverage crowns, the type of post was recorded as no post, a fiber post, or a metal post.

The incidence of fractures and their restorability were identified, and fractures were classified as restorable or unrestorable. Survival was defined using the following criteria: 1) survival: no fracture or a restorable fracture; 2) non-survival: an unrestorable fracture.

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Survival of restorati

CONCLUSIONS

The overall fracture survival rate of molar ETT restored with full-coverage crowns was higher than that of ETT restored with resin composite restorations. The survival rates of ETT with resin composites and crowns were only significantly different during the period of 12–24 months. The long-term survival rates (24–60 months) were not significantly different. Molar ETT with tooth structure loss on only the occlusal surface could be successfully restored with resin composite restorations.

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